

Pythagoras Investigation 2018

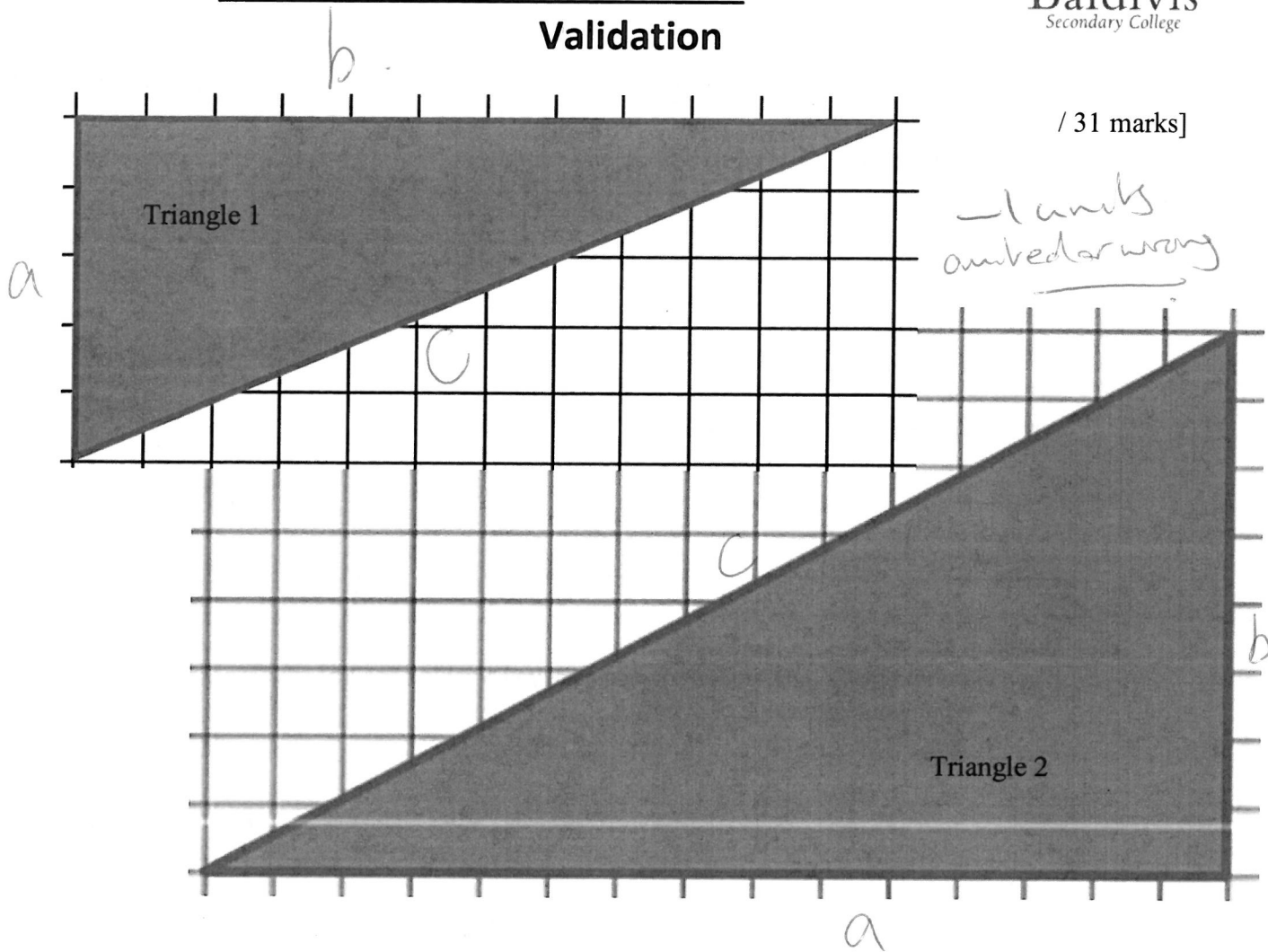


Baldyvis
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Name: _____

Validation

[31 marks]



Question 1.

[6 marks]

On the 2 triangles above, label the hypotenuse with the letter "c" and the other 2 sides with "a" and "b".

Question 2.

[6 marks]

For each of the 2 triangles above, measure the lengths of the sides and record those measurements in the table below.

Triangle	Side a	Side b	Side c
1	15	8	17
2	5	12	13

Question 3

[8 marks]

For both triangles complete the table below

Triangle	Area of square made from side $a =$ $a \times a = a^2$	Area of square made from side $b =$ $b \times b = b^2$	$a^2 + b^2$	Area of square made from side $c =$ $c \times c = c^2$
1	225	64	289	289
2	25	144	169	169

✓
✓✓
✓

✓

✓

Question 4.

[3 marks]

Write equation linking a^2 , b^2 and c^2 ?

$$\underline{a^2} + \underline{b^2} = \underline{c^2} \quad \text{or} \quad \begin{aligned} a^2 &= c^2 - b^2 \\ b^2 &= c^2 - a^2 \end{aligned}$$

Question 5.

[2 marks]

If you were told that a right angled triangle gave you a value of $a^2 + b^2$ equal to 9;a) What would the value of c^2 be?

9. ✓

b) Can you now work out the length of side "c"? If so what would it be?

3. ✓

Question 6

[2 marks]

In a right angled triangle, $a^2 = 49$ and $b^2 = 576$ a) What is the value of c^2 .

$$49 + 576 = 625$$

b) what is the length of side c?

25.

Question 7

[2 marks]

In a right angled triangle, side a = 12 cm and side b = 9cm.

- a) What is the value of c^2 .

$$12^2 + 9^2 = 144 + 81 = 225 \text{ cm}^2$$

- b) what is the length of side c?

15cm

Question 8.

[2 marks]

If you were told that $c^2 = 169$ and that $b^2 = 144$;

- a) What would the value of a^2 be?

~~81~~ 25

- b) Can you work out the length of side "a"? If so what is it?

5

End of Investigation

